

SAHAR JAVADI

Neuroscientist | Postdoctoral Researcher | Drug Discovery & CNS

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SUMMARY

- Neuroscientist with extensive experience in academia and industry, leading projects in neurodevelopmental and neurodegenerative diseases with 6+ first-author/co-author publications and manuscripts.
- Established rescue effects of repurposed therapeutics for neurodevelopmental and neurodegenerative disorders.
- Validated and optimized cell-based and plate-based assays for Janssen (Johnson & Johnson) pipelines in CNS and oncology.
- Contributed to mechanistic studies identifying how exercise enhances adult hippocampal neurogenesis.
- Led candidate gene functional validation (2 genes) for neuronal resilience in tauopathy using 2D and 3D iPSC-derived models.
- Experienced educator: taught and mentored 500+ undergraduate students and supervised 10+ undergraduate researchers.
- Multilingual (4 languages) with cross-cultural experience working in multiple countries.

TECHNICAL SKILLS

- In Vivo Models: Rodent behavioral assays, colony management, breeding strategy design, IP and SC injections, blood/tissue collection, perfusion, brain dissection, and preparation of samples for histological and molecular analysis.
- Cellular Models: iPSC, NPC, neurons, astrocytes, 2D/3D cultures, primary cells.
- Cellular Assays: Proliferation, differentiation, maturation, viability, apoptosis, necrosis, mitochondrial/oxidative stress.
- Molecular Biology: Genotyping, qPCR, Western blot, DNA/RNA/protein isolation and qualification/quantification, ICC/IHC, shRNA screening, lentiviral production, purification, cell transfection and transduction.
- Imaging: Light/fluorescence/confocal microscopy, high-content imaging, multiplex IF staining, FFPE tissue handling and analysis
- Lab Automation and Digital Documentation: Hamilton STAR, Echo650, BioTek FX, eLabJournal, BioVia.
- Data Analysis and Software: Microsoft Office, GraphPad Prism, Stereo Investigator, BioRender, SnapGene, R, ImageJ, ImageXpress, omics analysis, pathway enrichment analysis.
- Expanding expertise in AI/ML applications for neuroscience, including model-guided analysis of imaging, omics datasets, and neuronal phenotyping.

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar | UCLA Department of Neurology | 2023–Present

- Led functional screening of two candidate neuronal resilience genes, **HIF2A** and **NFE2L1**, across 2D and 3D iPSC-derived neuron models, identifying them as a robust driver of stress-resilience phenotypes.
- Co-led a drug repurposing program targeting the HIF2A resilience axis in iPSC-derived neurons and a tauopathy mouse model, validating 3 lead candidates via proteomics and immunohistochemistry, in partnership with the campus proteomics core and a collaborating mouse behavior lab.
- Led a systematic, multi-modal study of tau mutation-driven splicing impairment, integrating long-read sequencing to resolve isoform-level mis-splicing, global proteomics to map downstream protein-level consequences, and EV proteomics from neuron-conditioned media to evaluate biomarker potential, establishing an end-to-end discovery pipeline from molecular mechanism to translatable biomarker candidate.

- Modeled aging-associated stress pathways (oxidative stress, mitochondrial damage, DNA damage) to map their convergence with tauopathy, generating mechanistic evidence linking HIF2A activity to reduced tau pathology that informed a follow-up R01 grant application.
- Characterized 3 tauopathy iPSC derived neuronal models using multi-omics profiling, establishing a transcriptomic/proteomic resource now used by 3 other lab members for downstream target validation studies.

Assay Development Scientist | Eurofins Contract @ Janssen Pharmaceuticals (Johnson & Johnson) | 2022–2023

- Validated and optimized cell-based and biochemical assays for 10+ small and large molecules in CNS and cancer pipelines.
- Collaborated with the characterization department to analyze product impurities using protein analytical methods.
- Authored SOPs, assay validation reports, and technical documentation aligned with J&J quality standards.
- Interpreted complex biological datasets and presented results to cross-functional teams.

Ph.D. Researcher & Teaching Assistant | University of Wisconsin–Madison | 2015–2022

- First-author study: Repurposed FDA-approved drug Nutlin-3 demonstrated to rescue neurogenesis and cognition long-term in Fragile X syndrome (Fmr1 KO) mice, identifying ECM- and niche-mediated mechanisms.
- Performed aNPC culture, neuronal/astrocyte differentiation, and proteomics validations of muscle-derived factors regulating neural fate (co-author).
- Conducted viral transduction, lentiviral production, shRNA screening, and molecular cloning; led in vitro NPC proliferation/neuronal differentiation assays to study RGS6 in adult-born neuronal fate (co-author).
- Led analysis of age-dependent neurogenesis changes in Fragile X models (in vitro and in vivo); published.
- Characterized neurogenesis and brain morphology in a Disc1 rat model of schizophrenia for downstream stress/genetics interaction analysis for neuropsychiatric conditions.
- Taught and supervised 500+ undergraduate students in genetics, physiology, molecular biology, and microscopy as part of the Integrative Biology curriculum at UW–Madison.
- Awarded the Distinguished Graduate Student Fellowship (2019).

Breeding Program Scientist | Business Development Specialist | Amard Dam Tabarestan Animal Breeding, Iran | 2013–2015

- Designed and implemented population genetics–based breeding strategies, optimizing selection for productivity, fertility, longevity, and disease resistance traits.
- Developed data-driven performance tracking systems to monitor herd productivity metrics, reproductive outcomes, and longitudinal genetic trends.
- Applied quantitative genetics and statistical modeling approaches to improve selection efficiency and long-term population sustainability.
- Supported strategic business development initiatives through client engagement, breeding program expansion, partnership discussions, and commercialization planning.

• EDUCATION

- **Ph.D., Neuroscience — University of Wisconsin–Madison**
- **M.Sc., Animal Genetics — University of Tehran**
- **B.Sc., Animal Science — University of Guilan**

SELECTED PUBLICATION

- **Javadi, S.**, Li Y, Sheng J, Zhao L, Fu Y, Wang D, Zhao X. "Sustained correction of hippocampal neurogenic and cognitive deficits after a brief treatment by Nutlin-3 in a mouse model of fragile X syndrome." *BMC Medicine*, 20(1):163 (2022).
- **Javadi, S.**, Kuhu G, Raina S, Jessica R. "NFE2L1-Dependent Proteostasis Supports Neuronal Resilience in Human iPSC-Derived Models of Tauopathy." *In progress* (2026).
- **Javadi, S.**, Bilousova T, Sen M, Jessica R. "Mutation-Specific Disruption of Tau Pre-mRNA Splicing in iPSC Neurons and Its Signature in Extracellular Vesicles." *In progress* (2026).
- Gao Y, Shen M, Gonzalez JC, Dong Q, Kannan S, Hoang JT, Eisinger BE, ... **Javadi S.**, et al. "RGS6 mediates effects of voluntary running on adult hippocampal neurogenesis." *Cell Reports*, 32(5):107997 (2020).
- Moon HY, **Javadi S.**, Stremlau M, Yoon KJ, Becker B, Kang SU, Zhao X, et al. "Conditioned media from AICAR-treated skeletal muscle cells increases neuronal differentiation of adult neural progenitor cells." *Neuropharmacology*, 145:123-130 (2019).